AI Study Worksheet

1. Agents & Problem Formulation

Multiple Choice Questions

1. Which of the following is **not** a property of knowledge-based agents?
a) They can adapt by updating knowledge.
b) They can derive new conclusions using inference.
c) They only operate with fixed, pre-programmed behaviors.
d) They accept tasks in the form of goals.

2. In a search problem, the **transition model** specifies:
a) The starting state of the agent.
b) The set of all possible goal states.
c) The result of applying an action to a state.
d) The cost associated with reaching a state.

Open-ended

- Explain the difference between the declarative and procedural approaches to building agents. Why might combining both be beneficial?

2. Search Strategies

Multiple Choice Questions

1. Depth-first search (DFS) is most appropriate when:
a) The graph is infinite.
b) Memory is severely limited.
c) The shortest path is required.
d) Heuristic information is available.

2. Which condition ensures that A* search finds an optimal solution?
a) The heuristic is consistent.
b) The branching factor is small.
c) The frontier is treated as a stack.
d) The search space is acyclic.

Open-ended

- Compare the time and space complexity of DFS and BFS. Under what conditions would you prefer each?
- What does it mean for a heuristic to be admissible? Give an example.

3. Logic & Inference

Multiple Choice Questions

1. Which inference rule allows us to derive β from $(\alpha \to \beta)$ and α?
a) And-elimination
b) Modus ponens
c) Resolution
d) Biconditional elimination

2. Which property guarantees that an inference procedure will find all valid conclusions?
a) Soundness
b) Completeness
c) Consistency
d) Efficiency

Open-ended

- Give an example of how De Morgan's Law rewrites a logical expression. Why are such equivalences important in automated reasoning?
- Explain the role of resolution in theorem proving. How does converting formulas to CNF help?

4. Constraint Satisfaction Problems (CSPs)

Multiple Choice Questions

1. The AC-3 algorithm enforces:
a) Global consistency
b) Node consistency
c) Arc consistency
d) Path consistency

2. What is the worst-case time complexity of enforcing GAC on a CSP with e arcs and domain size d?
a) $O(ed^2)$
b) $O(ed^3)$
c) $O(d^n)$

d) $O(n \cdot d)$

Open-ended

- Explain the difference between hard and soft constraints with a real-world example.
- Why is AC3 often not enough to find a solution? What are the possible outcomes of running this algorithm?

5. Local Search & Optimization

Multiple Choice Questions

1. Which issue does simulated annealing help address compared to hill climbing?
a) High branching factor
b) Getting stuck in local optima
c) Lack of evaluation function
d) Infinite search trees

2. Which of these is a population-based local search method?
a) Hill climbing
b) Depth-first search
c) Beam search
d) Genetic algorithms

Open-ended

- Compare random restart hill climbing and simulated annealing. In what situations might one outperform the other?
- How do genetic algorithms combine exploration and exploitation? Illustrate with the role of crossover and mutation.

6. Monte Carlo Tree Search & AlphaGo

Multiple Choice Questions

1. Which of the following is **not** a step in MCTS?
a) Selection
b) Expansion
c) Simulation
d) Compilation

2. Why is brute-force game tree search in Go intractable?
a) The branching factor is too small.
b) The evaluation function is too simple.

c) The search tree size exceeds the number of atoms in the universe.
d) The game has no defined end.

Open-ended

- Describe how the policy network and value network improved MCTS in AlphaGo.
- What role does the multi-armed bandit problem play in balancing exploration and exploitation in MCTS?